

Diabetes Therapy Landscape

A Comprehensive Overview of Pharmaceutical, Technological
and Emerging Therapeutic Approaches

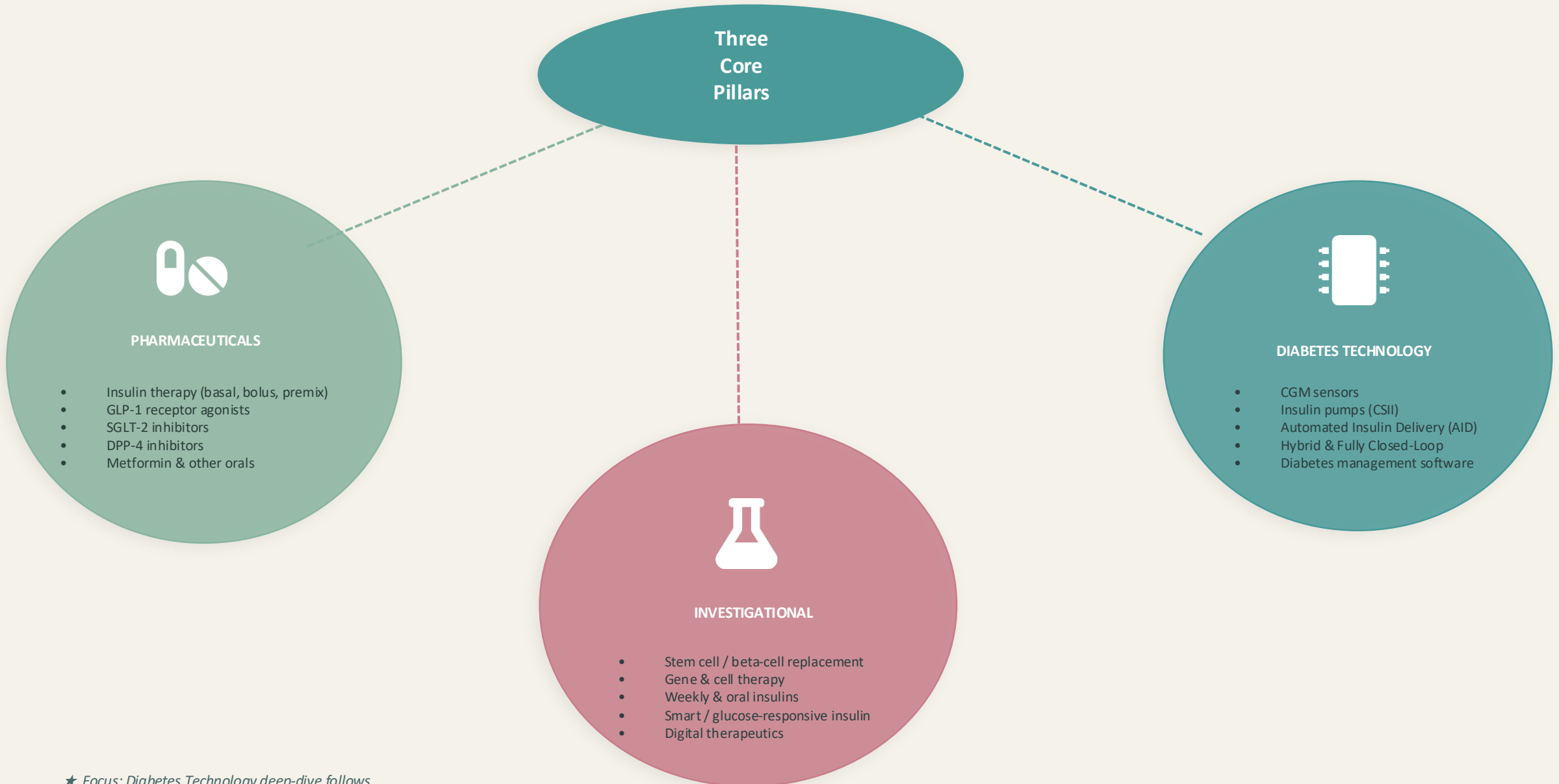
Pharmaceuticals

Diabetes Technology

Investigational & Supportive

FDA & EMA Regulatory References | 2024–2025

Diabetes Therapy Landscape



★ Focus: Diabetes Technology deep-dive follows

Pharmaceuticals Overview

Insulins

Basal | Bolus | Premix | Ultra-long-acting

Indication: T1D & T2D

Glargine (Lantus/Toujeo)
Degludec (Tresiba)
Lispro (Humalog)
Aspart (NovoRapid)

GLP-1 RAs

Receptor Agonists — injectables & orals

Indication: T2D (+ obesity)

Semaglutide (Ozempic/Wegovy/Rybelsus)
Liraglutide (Victoza)
Tirzepatide (Mounjaro)

SGLT-2 Inhibitors

Sodium-glucose co-transporter 2

Indication: T2D; T1D (EMA)

Empagliflozin (Jardiance)
Dapagliflozin (Farxiga)
Canagliflozin (Invokana)

DPP-4 Inhibitors

Dipeptidyl peptidase-4

Indication: T2D

Sitagliptin (Januvia)
Saxagliptin (Onglyza),
Alogliptin (Nesina)

Biguanides & Others

Metformin, TZDs, meglitinides, SUs

Indication: T2D

Metformin (first-line T2D)
Pioglitazone
Glipizide

Amylin Analogues

Adjunct to mealtime insulin

Indication: T1D & T2D

Pramlintide (SymlinPen)

Diabetes Technology: Evolution of Insulin Delivery

1920s–1990s

MDI

Multiple Daily Injections

- Manual basal + bolus injections
- Insulin pens & syringes
- SMBG finger-stick monitoring
- No automation

1

2

1970s–present

CSII

Continuous Subcutaneous Insulin Infusion

- Programmable basal rates
- On-demand bolus
- Infusion set / cannula
- CGM or SMBG required

2014–present

Hybrid Closed-Loop

AID / HCL — CGM + Pump + Algorithm

- Automated basal adjustment
- Manual mealtime bolus still needed
- Predictive low glucose suspend
- Multiple FDA & EMA approvals

3

4

2022–present

Fully Closed-Loop

& DIY / Open-source AID

- Automated bolusing (advancing)
- iLet Bionic Pancreas (FDA-cleared)
- CamAPS FX (EMA, FCL modes)
- DIY: OpenAPS, Loop, AAPS

Insulin Pumps on Market : Indications & Regulatory Status

Device	Type	Indication	Age	FDA	EMA	AID System	CGM
MiniMed 780G (Medtronic)	Tubed	T1D	≥7 yrs	✓	✓	HCL — SmartGuard	Guardian Sensor 3/4
MiniMed 670G (Medtronic)	Tubed	T1D	≥2 yrs	✓	✓	HCL (1st FDA AID)	Guardian Sensor 3
t:slim X2 (Tandem)	Tubed	T1D & T2D*	≥6 yrs	✓	✓	HCL — Control-IQ	Dexcom G6, G7
Omnipod 5 (Insulet)	Tubeless	T1D	≥2 yrs	✓	✓	HCL — SmartAdjust™	Dexcom G6
Omnipod DASH (Insulet)	Tubeless	T1D & T2D	≥2 yrs	✓	✓	No (stand-alone)	No
CamAPS FX (CamDiab)	App+pump	T1D, T2D, preg.	≥2 yrs	No	✓	HCL / FCL modes	Dexcom G6
iLet Bionic Pancreas	Tubed	T1D	≥6 yrs	✓	No	FCL — insulin+glucagon	Dexcom G6
mylife YpsoPump (Ypsomed)	Tubed	T1D, preg.	All	No	✓	HCL (CamAPS FX)	Dexcom G6
Dana-i / RS (SOOIL)	Tubed	T1D & T2D	All	No	✓	DIY (Loop/AAPS)	Multiple
ACCU-CHEK Solo (Roche)	Tubeless	T1D & T2D	All	No	✓	No (stand-alone)	No

*T2D varies by jurisdiction. Data as of 2024–2025.

CGM Sensors : Market Overview & Pump Pairings

Sensor	Wear	Calibration	Indication / Age	FDA	EMA	AID-Compatible Pumps
Dexcom						
G6	10 days	Factory cal.	T1D, T2D, ≥2 yrs	✓	✓	t:slim X2, Omnipod 5, iLet, CamAPS FX
G7	10d + 12h grace	Factory cal.	T1D, T2D, ≥2 yrs	✓	✓	t:slim X2 (latest SW), expanding
Stelo (OTC)	15 days	Factory cal.	T2D (non-insulin)	✓	No	Consumer / stand-alone
Abbott						
FreeStyle Libre 2	14 days	Factory cal.	T1D, T2D, ≥4 yrs	✓	✓	No certified AID integration (iCGM)
FreeStyle Libre 3	14 days	Factory cal.	T1D, T2D, ≥4 yrs	✓	✓	Expanding; Libre 3 Plus in US
FreeStyle Libre 2 Plus	15 days	Factory cal.	T1D, T2D	✓	✓	Omnipod 5 (rolling out)
Medtronic						
Guardian Sensor 3	7 days	2×/day fingerstick	T1D, ≥7 yrs (pump)	✓	✓	MiniMed 670G, 770G, 780G
Guardian Sensor 4	7 days	Factory cal.	T1D, ≥7 yrs	✓	✓	MiniMed 780G exclusively
Senseonics						
Eversense 365	365 days (implant)	1×/day tap	T1D & T2D, ≥18 yrs	✓	✓ (E3)	No certified AID integration

AID Systems: Algorithms, Software & Data Platforms

COMMERCIAL AID ALGORITHMS

1

Control-IQ (Tandem) [\[HCL\]](#)
Predictive algorithm targeting 112.5 mg/dL. Sleep & Activity modes. Control-IQ+ adds T2D indication.
SW: t:connect, Tandem Source

2

SmartAdjust™ (Insulet) [\[HCL\]](#)
Omnipod 5. 10-min predictive adjustments on phone or pod controller. No-announce meal detection.
SW: Omnipod 5 App, Glooko

3

SmartGuard™ (Medtronic) [\[HCL\]](#)
MiniMed 780G. MPC algorithm, auto-corrects every 5 min. Targets 100 or 110 mg/dL.
SW: Guardian 4 app, CareLink

4

CamAPS FX (CamDiab) [\[HCL/FCL\]](#)
Cambridge adaptive MPC. Android. Unique approvals: pregnancy & paediatrics ≥1 yr. FCL mode.
SW: CamAPS FX (Android)

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iLet Algorithm (Beta Bionics) [\[FCL\]](#)
Fully autonomous. Bihormonal (insulin + glucagon). Body weight only required input. No carb counting.
SW: iLet app

DIY / OPEN-SOURCE SYSTEMS



OpenAPS Dana, Roche Combo + CGM
Original open-source AID; foundational algorithm for many commercial systems



Loop (iOS) Omnipod Eros/DASH, Medtronic + Dexcom/Libre
iPhone-based. Loop 3.x. Nightscout-free option available



AndroidAPS (AAPS) Dana-i/RS, ACCU-CHEK, Eopatch + multi CGMs
Android. Oref0/Oref1/SMB. Widely used across Europe



Tidepool Loop Omnipod 5 (FDA-cleared)
First FDA-cleared open-source DIY AID system. Dexcom G6

DATA MANAGEMENT PLATFORMS

Tidepool: Open-source. Aggregates CGM, pump, meter data. HCP + patient dashboards.

Glooko: Clinic-facing. Integrates Omnipod 5, CGMs, meters. FDA 510(k) cleared.

CareLink (Medtronic): MiniMed + Guardian CGM platform. Remote HCP monitoring.

Dexcom Clarity: Time-in-range analytics. EHR integration. Patient & clinic portals.

Abbott LibreView: AGP reports. EHR connectivity. Pharmacy integrations.

Nightscout: Open-source remote CGM monitoring. Widely used with DIY AID.



Investigational & Supportive Therapies : Horizon Scanning



BETA-CELL RESTORATION

- Stem cell-derived islet transplantation (Vertex VX-880 / VX-264 — Phase 1/2)
- Encapsulated islet cells (immune rejection prevention)
- Islet allograft transplantation (donor-limited)
- In-vivo beta-cell regeneration (GLP-1 + gastrin)



GENE & CELL THERAPY

- AAV-based gene therapy for insulin gene expression
- CRISPR editing for T1D immune tolerance
- Precision medicine - monogenic diabetes (MODY)
- Teplizumab (Tzield) - delays T1D onset (FDA 2022)



NEXT-GEN INSULIN DELIVERY

- Oral insulin formulations (Oramed OLP-114 - Phase 3)
- Weekly insulins: Icodec (Awiqli, FDA & EMA 2023/24)
- Smart (glucose-responsive) insulin - GRI (early clinical)
- Patch-based micro-dosing systems (wearable, needle-free)



DIGITAL & SUPPORTIVE

- FDA-cleared digital therapeutics for T2D (Livongo, One Drop)
- Continuous ketone monitoring (CKM) - T1D DKA risk
- Non-invasive CGM (optical, microneedle (research stage))
- Dual hormone artificial pancreas (insulin + glucagon/amylin)

Key Takeaways

01

Pharmacotherapy remains foundational

Insulin therapy underpins T1D; GLP-1 RAs and SGLT-2 inhibitors are reshaping T2D with expanding CV and renal indications.

02

AID systems are the new standard of care

Multiple FDA- and EMA-approved HCL systems span tubed and tubeless formats. The field is advancing rapidly toward fully closed-loop.

03

CGM is the enabling technology

Factory-calibrated 10–15 day sensors are table stakes. Implantable 365-day CGM and OTC options are expanding access.

04

Algorithms & software are the differentiators

Control-IQ, SmartGuard, CamAPS FX, and open-source DIY ecosystems compete on algorithm sophistication and data platform depth.

05

The horizon includes disease modification

Teplizumab, stem cell approaches (Vertex), weekly insulins, and glucose-responsive insulin represent the next frontier.